



PUNJAB PUBLIC SERVICE COMMISSION
COMBINED COMPETITIVE EXAMINATION
FOR RECRUITMENT TO THE POSTS OF
PROVINCIAL MANAGEMENT SERVICE, ETC -2021
CASE NO. 3C2022

SUBJECT: MATHEMATICS (PAPER-I)

TIME ALLOWED: THREE HOURS

MAXIMUM MARKS: 100

NOTE:

- i. All the parts (if any) of each Question must be attempted at one place instead of at different places.
- ii. Write Q. No. in the Answer Book in accordance with Q. No. in the Q. Paper.
- iii. No Page/Space be left blank between the answers. All the blank pages of Answer Book must be crossed.
- iv. Extra attempt of any question or any part of the question will not be considered.

NOTE: Attempt FIVE Questions in All. THREE Questions from Section 'A' and TWO Questions from Section 'B'. Calculator is allowed. (Not programmable)

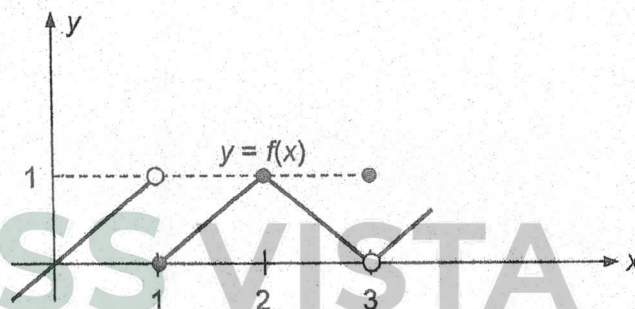
SECTION-A

Q.1 (a) For the function $f(x)$ graphed in the adjoining figure, find the following limits or explain why they do not exist.

(i) $\lim_{x \rightarrow 1} f(x)$ (ii) $\lim_{x \rightarrow 2} f(x)$

(iii) $\lim_{x \rightarrow 3} f(x)$

Also discuss the continuity of $f(x)$ at $x = 1$, $x = 2$ and $x = 3$.



(b) Differentiate $y = \sqrt[3]{\frac{x(x^2+1)}{(x-1)^2}}$ with respect to x .

(10 + 10 = 20 Marks)

Q.2 (a) For what values of a , m , and b does the function

$$f(x) = \begin{cases} 3, & x = 0 \\ -x^3 + 3x + a, & 0 < x < 1 \\ mx + b, & 1 \leq x \leq 2 \end{cases}$$

satisfy the hypotheses of mean value theorem on the interval $[0, 2]$?

(b) A box with rectangular base, whose length is twice its width, is to have a closed top. The area of the material in the box is to be 192 in^2 . What should the dimensions of the box be in order to have the largest possible volume?

(10 + 10 = 20 Marks)

Q.3 (a) Show that $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \frac{\pi}{4}$.

(b) Evaluate $I = \int_0^4 \int_0^{4-x} \int_0^{4-x-y} dz dy dx$. (10 + 10 = 20 Marks)

- Q.4 (a) A plot of land lies between a straight fence and a curved stream at distance x meter from one end of the fence, the width y meters of the plot was measured as follows

x	0	10	20	30	40	50	60	70	80
y	0	32	44	58	63	50	30	32	0

Find the approximate area of the plot by using trapezoidal rule.

- (b) Solve the differential equation $(x+1)\frac{dy}{dx} - ny = e^x(x+1)^{n+1}$.

(10 + 10 = 20 Marks)

- Q.5 (a) Assume that the half life of the radium in a piece of lead is 1500 years. How much radium will remain in the lead after 2500 years?

- (b) A particle of mass m is moving under the action of the forces $F_1 = -m\omega^2 x$, $F_2 = mF_0 t$, $F_3 = -2m\mu \frac{dx}{dt}$. Assuming that damping is small, set up and solve the equation of motion.

(10 + 10 = 20 Marks)

SECTION-B

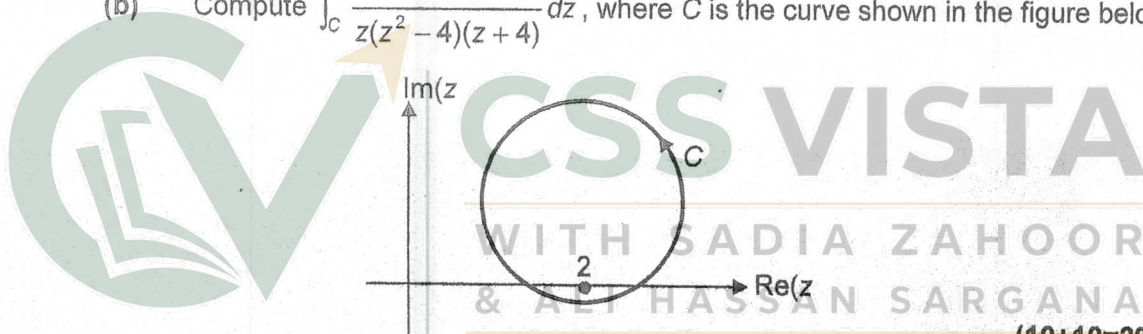
- Q.6 (a) Find radius and interval of convergence of the series of $\sum_{n=1}^{\infty} \frac{n! x^n}{(2n)!}$.

- (b) Prove that $(\sin x + i \cos x)^n = \cos n\left(\frac{\pi}{2} - x\right) + i \sin n\left(\frac{\pi}{2} - x\right), n \in \mathbb{Z}$.

(10 + 10 = 20 Marks)

- Q.7 (a) Construct the analytic function whose real part is $e^{-x}[(x^2 - y^2) \cos y + 2xy \sin y]$.

- (b) Compute $\int_C \frac{z^2 + 2}{z(z^2 - 4)(z + 4)} dz$, where C is the curve shown in the figure below:



(10+10=20 Marks)

- Q.8 (a) Find the tangent and normal to the curve $x^2 - xy + y^2 = 7$ at the point $(-1, 2)$.

- (b) Find the curvature and torsion of the circular helix $\vec{r} = (a \cos u, a \sin u, bu)$.

(10 + 10 = 20 Marks)



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CASE NO. 3C2022

SUBJECT: MATHEMATICS (PAPER-II)

TIME ALLOWED: THREE HOURS

MAXIMUM MARKS: 100

NOTE:

- i. All the parts (if any) of each Question must be attempted at one place instead of at different places.
- ii. Write Q. No. in the Answer Book in accordance with Q. No. in the Q. Paper.
- iii. No Page/Space be left blank between the answers. All the blank pages of Answer Book must be crossed.
- iv. Extra attempt of any question or any part of the question will not be considered.

NOTE: Attempt any five questions. All questions carry equal marks. Calculator is not allowed (Programmable)

Q.No.1

- a) Let P_2 be a set of all polynomials having degree at most equal to 2 and $T: P_2 \rightarrow P_2$ a linear transformation such that $T(3 - 5x) = 2$ and $T(1 - x + 2x^2) = 1 + x$. Find $T(6 - 11x - 3x^2)$.
- b) Let V and W be two vector spaces over the same field F and $T: V \rightarrow W$ a linear transformation. Show that T is one-to-one if and only if $\text{Ker}(T) = \{0\}$.

(10 + 10 = 20 Marks)

Q.No.2

- a) Find the eigen values and the corresponding eigen vectors of $A = \begin{bmatrix} -1 & 0 & 1 \\ 3 & 0 & -3 \\ 1 & 0 & -1 \end{bmatrix}$.
- b) If possible, find a matrix P such that $P^{-1} \begin{pmatrix} 1 & 3 \\ 2 & 2 \end{pmatrix} P$ is a diagonal matrix.

(10 + 10 = 20 Marks)

Q.No.3

- a) Using Gram-Schmidt process, orthonormalize the set $\{1, t, t^2\}$ on the interval $[-1, 1]$ with the following inner product $\langle x, y \rangle = \int_{-1}^1 x(t) \cdot y(t) dt$.
- b) Show that $C[a, b] = \{f: [a, b] \rightarrow \mathbb{R} \text{ where } f \text{ is a continuous function}\}$ is not an inner product space.

(10 + 10 = 20 Marks)

Q.No.4

- a) Let (X_1, d_1) and (X_2, d_2) be two metric spaces. If (X_1, d_1) is a complete metric space and isometric with (X_2, d_2) , show the (X_2, d_2) is a complete metric space.
- b) Let (X, d) be a metric space and T a collection of all subsets of X . Show that T is a topology on X .
Recall that, a set U in X is open if for each $x \in U$, there exists a real number $r > 0$ such that $x \in S_r(x) \subseteq U$.

(10 + 10 = 20 Marks)

Q.No.5

- a) If H is a non-empty finite subset of a group G and H is closed under multiplication, then show that H is a subgroup of G .
 - b) If G is a finite group and $g \in G$, then show that order of g divides the order of G .
- (10 + 10 = 20 Marks)**

Q.No.6

- a) If ϕ is a homomorphism of group G_1 into a group G_2 with kernel K , then by proving $\phi(x^{-1}) = [\phi(x)]^{-1}$, show that K is a normal subgroup of G .
 - b) If G is a group, then show that the set of all automorphisms of G is a group.
- (10 + 10 = 20 Marks)**

Q.No.7

- a) Show that a finite integral domain is a Field.
 - b) If $v_1, v_2, \dots, v_n$ are vectors in a vector space V , then show that either they are linearly independent or some Vector v_m is a linear combination of $v_1, v_2, \dots, v_{m-1}$.
- (10 + 10 = 20 Marks)**

Q.No.8

- a) Find a basis and dimension of the span of $\{[1, 4, -1, 3], [2, 1, -3, -1], [0, 2, 1, -5]\}$
 - b) Give at least four examples of infinite dimensional vector spaces and show that the set of all polynomials in x with real coefficients is an infinite dimensional vector space.
- (10 + 10 = 20 Marks)**



CSS VISTA
WITH SADIA ZAHOOR
& ALI HASSAN SARGANA